

HUTCHINSON COUNTY AP, TX, USA

WMO: 723635

Lat: 35.695N

Lon: 101.395W

Elev: 3054

StdP: 13.14

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 03024

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.9	18.0	-0.5	6.0	33.2	2.8	7.1	35.7	28.7	50.7	25.8	52.2	8.9	10	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	100.8	67.1	98.8	67.4	96.5	67.2	72.6	88.8	71.4	88.0	70.4	87.3	11.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	116.5	78.0	66.7	110.2	77.1	65.7	106.3	76.5	38.3	88.7	37.1	87.7	36.2	87.4	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.9	23.2	20.3	4.8	105.5	4.2	2.9	1.8	107.6	-0.7	109.4	-3.0	111.0	-6.0	113.2	
			DB												
			WB	4.3	74.8	4.9	2.8	0.8	76.8	-2.1	78.4	-4.8	80.0	-8.3	82.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.9	40.0	42.7	51.6	59.9	68.7	78.4	82.6	81.0	73.4	61.3	50.0	40.1	(6)
		DBStd	17.65	10.02	11.23	10.29	9.44	8.87	6.33	5.22	5.63	7.15	9.19	9.99	9.96	
		HDD50	1185	335	250	105	25	0	0	0	0	19	120	329		
		HDD65	3515	776	624	428	203	65	1	0	15	177	454	772		
		CDD50	5165	24	47	154	321	582	853	1010	960	702	369	121	22	
		CDD65	2022	0	1	12	50	180	404	545	495	268	62	5	0	
		CDH74	24600	3	37	226	758	2201	4807	7028	5903	2821	712	102	2	
		CDH80	12747	0	7	59	279	1023	2566	4019	3224	1306	251	13	0	
(14)	Wind	WSAvg	10.4	9.6	10.3	11.0	12.4	11.2	11.4	10.1	9.4	10.0	9.9	9.9	9.6	(14)
(15)	Precipitation	PrecAvg	20.6	0.6	0.6	0.9	1.9	3.0	2.7	2.8	2.9	1.7	1.5	0.9	0.8	(15)
(16)		PrecMax	41.3	1.9	1.7	4.3	8.5	10.0	6.0	8.7	5.4	4.6	5.2	4.7	3.5	(16)
(17)		PrecMin	11.3	0.0	0.0	0.0	0.0	0.5	0.1	0.4	0.5	0.0	0.0	0.1	0.1	(17)
(18)		PrecStd	5.6	0.5	0.4	1.0	1.9	1.9	1.6	1.7	1.3	1.2	1.5	1.0	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	72.5	79.0	86.3	91.7	98.8	104.6	104.7	102.5	99.2	91.8	81.9	72.1		(19)
(20)		0.4% MCWB	48.9	50.5	54.7	56.6	61.7	63.4	67.4	67.9	65.7	60.7	54.2	48.8		(20)
(21)		2% DB	67.7	72.4	80.7	87.7	94.2	99.6	101.0	99.7	95.0	86.5	77.0	65.7		(21)
(22)		2% MCWB	46.5	48.9	53.7	55.9	60.3	65.8	67.9	68.1	65.3	60.7	53.1	46.8		(22)
(23)		5% DB	62.4	67.6	74.8	82.5	90.5	96.3	98.7	97.3	91.4	82.0	72.1	61.2		(23)
(24)		5% MCWB	44.5	47.3	51.4	55.3	61.2	66.8	68.4	68.3	65.7	59.4	51.8	44.8		(24)
(25)		10% DB	56.8	61.7	69.9	77.9	86.2	92.7	96.1	94.4	87.9	77.1	66.5	56.4		(25)
(26)		10% MCWB	42.0	44.0	50.3	54.7	60.9	67.0	68.4	68.2	65.2	58.1	49.5	42.6		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	50.8	54.3	60.6	64.4	69.5	74.1	74.8	74.2	71.3	68.4	59.5	52.9		(27)
(28)		0.4% MCDB	67.5	71.4	73.3	75.5	83.9	89.4	90.6	88.8	86.7	79.2	71.1	63.3		(28)
(29)		2% WB	47.9	50.4	57.5	61.6	67.7	72.0	72.7	72.3	69.6	65.5	56.5	48.5		(29)
(30)		2% MCDB	64.6	68.7	68.7	73.8	81.9	88.2	89.1	88.5	84.7	76.2	67.8	62.6		(30)
(31)		5% WB	45.3	48.0	54.6	59.2	66.0	70.7	71.3	71.1	68.4	63.4	54.1	45.9		(31)
(32)		5% MCDB	61.3	65.8	69.2	72.7	81.0	87.1	88.7	87.6	83.3	73.7	66.6	59.2		(32)
(33)		10% WB	42.5	45.4	51.8	57.0	64.2	69.5	70.3	70.1	67.3	61.0	51.5	43.2		(33)
(34)		10% MCDB	56.3	60.8	67.9	72.6	78.6	85.7	88.2	86.8	81.8	71.8	65.6	56.0		(34)
(35)	Mean Daily Temperature Range	MDBR	24.1	25.1	25.7	25.4	24.6	23.9	23.1	22.8	23.0	23.8	24.4	22.4		(35)
(36)		5% DB MCDBR	31.8	33.7	32.6	32.4	30.0	28.0	26.0	26.0	26.9	29.5	31.0	29.7		(36)
(37)		5% DB MCWBR	16.7	16.5	14.3	12.9	11.0	8.6	6.9	7.1	8.2	11.8	14.3	15.8		(37)
(38)		5% WB MCDBR	30.0	31.3	27.8	26.0	25.4	24.2	23.0	23.1	22.1	22.4	25.8	27.8		(38)
(39)		5% WB MCWBR	16.3	16.3	13.7	13.3	11.5	9.1	7.3	7.3	8.0	11.6	13.3	15.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.265	0.295	0.328	0.343	0.380	0.384	0.373	0.343	0.302	0.275	0.258		(40)
(41)		taud	2.596	2.585	2.501	2.419	2.437	2.380	2.405	2.404	2.467	2.566	2.566	2.592		(41)
(42)		Ebn at Noon	307	315	311	303	298	285	283	285	290	296	296	299		(42)
(43)		Edh at Noon	23	27	32	36	36	38	37	36	33	27	24	22		(43)
(44)	All-Sky Solar Radiation	RadAvg	966	1222	1613	1996	2223	2371	2299	2048	1753	1324	1027	836		(44)
(45)		RadStd	84	114	118	118	168	144	99	103	86	149	101	58		(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (17 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+176	N/A

Nomenclature: See separate page